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COMPUTER

CHAPTER 9B

# Secondary Memory

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EDITION October 2026

LANGUAGE Punjabi & English

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CHAPTER 9B

# Secondary Memory

## Computer Awareness

IN THIS CHAPTER

- Hard disks and SSDs
- Tracks and sectors
- Magnetic tape
- Optical discs
- Virtual memory
- Access methods and units

## SECONDARY MEMORY (AUXILIARY STORAGE)

Secondary Memory refers to storage devices that store data permanently. These devices may be built into the computer or connected externally.

Secondary Memory is also known as:

- Auxiliary Memory
- External Memory
- Backup Storage

Unlike Primary Memory (RAM), Secondary Memory retains data even when the power is switched off.

### Features of Secondary Memory

- Non-Volatile Memory
- Large Storage Capacity
- Permanent Data Storage
- Lower Cost Per GB
- Slower than Primary Memory
- CPU Cannot Access Directly

Important Exam Point

The CPU cannot directly access secondary memory. Data must first be transferred to RAM and then processed by the CPU.

## CLASSIFICATION OF SECONDARY MEMORY

- Secondary Memory
- Magnetic Storage or Magnetic Disk
- Hard Disk
- Floppy Disk
- Magnetic Tape

- ZIP Disk
- Solid State Storage
- SSD
- Pen Drive
- Memory Card
- Optical Storage
- CD
- DVD
- Blu-ray Disc
- HVD

## HARD DISK DRIVE (HDD)

Hard Disk Drive (HDD) is a magnetic storage device used to store large amounts of data permanently.

It is also called:

- Hard Disk
- Winchester Disk

Common Capacities

- 500 GB
- 1 TB
- 2 TB
- 4 TB

and above.

## STRUCTURE OF HARD DISK

A hard disk consists of several circular magnetic plates called Platters.

Data is stored on both sides of platters.

Speed of hard disk measured in revolution per minute

Main Components

- Platter
- Read/Write Head
- Actuator Arm
- Spindle
- Disk Controller Board

## DATA ORGANIZATION IN HARD DISK

- Platter
- Tracks
- Sectors

## Track

Circular paths on the disk surface where data is stored.

## Sector

Small subdivision of a track.

Important Exam Point

Traditionally: 1 Sector = 512 Bytes = $2^9$ Bytes

(Modern drives may use 4 KB sectors.)

## CYLINDER

Tracks having the same radius on all platters together form a Cylinder.

## FORMATTING

The process of dividing a disk into tracks and sectors is called Formatting.

Formatting helps the operating system locate and manage files efficiently.

## HARD DISK PERFORMANCE TERMS

1.Seek Time

Time taken by the read/write head to reach the required track.

Important Exam Point

Lower Seek Time = Faster Disk

2.Rotational Latency

Time taken by the disk to rotate the required sector under the read/write head.

3.Data Transfer Time

Time taken to transfer data from the disk to memory.

## HARD DISK INTERFACES

Hard disks connect to the motherboard through:

1.SATA

Serial Advanced Technology Attachment

Most commonly used interface. 2.PATA

Parallel Advanced Technology Attachment

Older technology.

3.SCSI

Small Computer System Interface

Used in servers and high-performance systems.

## HARD DISK ADDRESSING

Addressing refers to locating data on a disk.

1.CHS Addressing

Cylinder – Head – Sector Addressing

Also known as Disk Geometry.

2.LBA Addressing

Logical Block Addressing

Modern method used by operating systems.

Important Exam Point

LBA has largely replaced CHS addressing.

## SSD (SOLID STATE DRIVE)

It is a non-volatile storage device that uses Flash Memory instead of magnetic platters. ADVANTAGES OF SSD

- Faster than HDD
- No Moving Parts
- Silent Operation
- Lower Power Consumption
- More Reliable
- Shock Resistant

Important Exam Point

SSD is significantly faster than HDD because it has no moving mechanical parts.

## HDD VS SSD

| Feature | HDD | SSD |
|---|---|---|
| Technology | Magnetic | Flash Memory |
| Speed | Slower | Faster |
| Moving Parts | Yes | No |
| Noise | Present | Silent |
| Power Usage | Higher | Lower |
| Cost | Cheaper | Costlier |

# MAGNETIC TAPE

Magnetic Tape is a secondary storage device primarily used for:

- Data Backup
- Archival Storage
- Large Data Sets

Fritz Pfleumer patented magnetic recording tape in Germany in 1928. Magnetic tape entered computer data storage in the 1950s. Traditional tape sizes included quarter-inch and half-inch widths; lengths vary by format.

The surface of the tape is coated with iron oxide or chromium dioxide.

It records data by magnetization, can store and erase data, and can read the data several times.

FEATURES OF MAGNETIC TAPE

- Sequential Access
- Low Cost
- Large Capacity
- Used for Backup
- Non-Volatile

STRUCTURE OF MAGNETIC TAPE

Tape data is accessed sequentially along the recording medium; modern formats can locate files or blocks using metadata.

There are parallel tracks on the tape called tracks, and the number of parallel tracks present on the tape is where data is stored.

Traditional nine-track computer tape commonly stored eight data bits and a parity bit across its width. Modern tape formats use many more tracks and are not limited to EBCDIC.

In magnetic tape, data is called records. The character bit is stored as a pattern of magnetic spots across the width of the tape, and the magnetic spots are recorded on 8 tracks.

An extra bit called a parity bit is stored on the 9th track, called the parity track.

Data on the tape is stored in groups called data blocks,

each data block has a gap between them called an inter-block gap (IBG).

A vertical column in the tape is called a frame.

Inter-record gaps separate records on traditional tape. Their length depends on the recording format; 0.5 inch is not a universal value.

Tape widths depend on the format; examples include 1 inch, 1/2 inch and 1/4 inch.

Data is recorded on the tape in binary 0 or 1.

The device used to read or write on the tape is called a tape streamer.

Blocking is the process of grouping two or more records to form a block of data, and the number of records grouped in each block is called the blocking factor.

Data recorded density is the number of bytes of records per inch.

In all magnetic tape, data is read or written serially, not randomly.

Traditional tape drives use a supply (file) reel and a take-up reel. These reels move tape during both reading and writing; one reel is not exclusively for reading and the other for writing.

Inter-record gaps separate records by blank spaces, and inter-block gaps separate blocks by blank spaces.

TYPES OF MAGNETIC TAPE

- 1/2 Inch Tape Reel
- Tape Cartridge
- Streamer Tape
- Digital Audio Tape (DAT)

## IMPORTANT EXAM FACT

Magnetic Tape uses Sequential Access only.

## PEN DRIVE

Pen Drive is a portable flash memory device.

Other Names:

- USB Flash Drive
- Thumb Drive
- Jump Drive

FEATURES

- Plug and Play
- Portable
- Reusable
- Non-Volatile
- USB Interface

Common Capacity

2 GB to 1 TB and above.

## MEMORY CARD (SD CARD)

SD Stands for Secure Digital Card

Used in:

- Smartphones
- Cameras
- Tablets

Types

- Standard SD
- Mini SD
- Micro SD

## CD (COMPACT DISC)

Developed by Philips and Sony in 1982.

A standard CD is around 5 inches in diameter and 0.05 inches in thickness

Used for storing:

- Audio
- Video
- Software
- Documents

Capacity :Approximately 700 MB

## TYPES OF CD

CD-worm(write once read many)

CD-ROM :Read Only Memory

CD-R :Record Once

CD-RW :Rewritable Disc

## DVD (DIGITAL VERSATILE DISC)

DVD stores much more data than a CD.

DVD was co-developed by Sony, Panasonic, Philips, and Toshiba in 1995.

TYPES OF DVD

- DVD-ROM :Read Only
- DVD-R :Write Once
- DVD-RW :Rewritable
- BLU-RAY DISC (BD)

- Uses Blue Laser Technology.

- Capacity: 25 GB (Single Layer) ,50 GB (Dual Layer)

Important Correction

Blu-ray capacity is much larger than 2.5 GB.

- HVD :Holographic Versatile Disc

Proposed HVD capacities reached approximately 3.9 TB; this was a development target, not a typical commercially available disc.

HVD was proposed for very high-capacity optical storage and did not become a mainstream consumer format.

## FLASH MEMORY

Flash Memory is a non-volatile semiconductor memory.

Data remains stored even after power is switched off.

USES OF FLASH MEMORY

- SSD
- Pen Drive
- Memory Cards
- Smartphones
- Digital Cameras
- Set Top Boxes

ADVANTAGES

- Fast Access
- Reliable
- Portable
- Low Power Consumption

## VIRTUAL MEMORY

Virtual memory gives programs an address space that can exceed physical RAM. Paging may move less-used data to disk or SSD; this storage does not become physical RAM.

When RAM becomes full, the operating system moves less-used data from RAM to storage.

Important Exam Point

Virtual Memory increases apparent memory but is slower than RAM.

## MEMORY ACCESS METHODS

Memory Access refers to the process by which a computer system reads data from or writes data to its memory. Memory access is a fundamental operation in computing, as it allows the processor to retrieve instructions and data needed to execute programs. There are two main types of memory access:

1.Read: Fetching data or instructions from memory.

2.Write : Storing data or results into memory.

## These are 4 types of memory access methods:

1. SEQUENTIAL ACCESS

Data is accessed in linear sequential manner order.

Example:Magnetic Tape

2. RANDOM ACCESS

Any memory location can be accessed directly. Examples:RAM, ROM

3. DIRECT ACCESS

Combination of sequential and random access.

Examples:

- Hard Disk
- Optical Disk
- Flash Drive

4. ASSOCIATIVE ACCESS

Data is accessed by content rather than address.

Example:

Cache Memory

# MEMORY UNITS

Memory units are used to measure and represent data. These units, which can be measured in bits and bytes, offer a technique to estimate the volume of data which can be stored in a specific device or system. Some of the commonly used memory units are:

## Bit

Smallest unit of data.

1 Bit = 0 or 1

## Nibble

1 Nibble = 4 Bits

## Byte

1 Byte = 8 Bits

## Kilobyte (KB)

1 KB = 1024 Bytes (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)

## Megabyte (MB)

1 MB = 1024 KB (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)

## Gigabyte (GB)

1 GB = 1024 MB (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)

## Terabyte (TB)

1 TB = 1024 GB (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)

## Petabyte (PB)

1 PB = 1024 TB (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)

## Exabyte (EB)

1 EB = 1024 PB (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)

## Zettabyte (ZB)

1 ZB = 1024 EB (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)

## MEMORY SIZES

Register :bytes

Cache memory :kb/mb

RAM:GB

USB:GB

Magnetic disk/hard disk:TB

Magnetic tapes:PB

## QUICK REVISION NOTES

- Secondary Memory is non-volatile.
- HDD is magnetic storage.
- SSD uses flash memory.
- HDD contains tracks and sectors.
- Cylinder = Tracks of same radius.
- Formatting divides disk into tracks and sectors.
- Magnetic Tape uses sequential access.
- Pen Drive uses flash memory.
- Blu-ray uses blue laser.
- Virtual Memory uses hard disk space.
- Cache uses associative access.

- RAM uses random access.

## MOST IMPORTANT FACTS FOR COMPETITIVE EXAMS

1.HDD is also called Winchester Disk.

2.SSD has no moving parts.

3.1 Sector = 512 Bytes (traditional).

4.CHS = Cylinder Head Sector.

5.LBA = Logical Block Addressing.

6.Magnetic Tape uses Sequential Access.

7.Cache Memory uses Associative Access.

8.RAM uses Random Access. 9.DVD stores more data than CD.

10.Flash Memory is used in SSDs and Pen Drives.

## ONE-LINE REVISION POINTS

- HDD stores data magnetically.
- SSD uses flash memory.
- Cylinder = Tracks of same radius.
- Seek Time affects HDD speed.
- SATA is the most common HDD interface.
- Magnetic Tape is best for backup.
- Pen Drive is a portable flash storage device.
- SD stands for Secure Digital.
- Blu-ray uses blue laser technology.
- Virtual memory can use disk space as backing storage for memory pages.
- Sequential Access = Tape.
- Random Access = RAM.
- Associative Access = Cache.
- 1 Byte = 8 Bits.
- 1 TB = 1024 GB (traditional binary exam convention; standardized binary prefixes use KiB, MiB, etc.)
- Memory Types Speed Order:

Register → Cache → RAM → SSD → HDD → Tape

(Fastest → Slowest)

## Binary Memory Units from the Source Diagram

| Unit | Binary size |
|---|---|
| Kibibyte (KiB) | $2^{10}$ bytes |
| Mebibyte (MiB) | $2^{20}$ bytes |
| Gibibyte (GiB) | $2^{30}$ bytes |
| Tebibyte (TiB) | $2^{40}$ bytes |
| Pebibyte (PiB) | $2^{50}$ bytes |
| Exbibyte (EiB) | $2^{60}$ bytes |
| Zebibyte (ZiB) | $2^{70}$ bytes |
| Yobibyte (YiB) | $2^{80}$ bytes |
| Brontobyte / geopbyte | Informal names in the source diagram, not standardized SI or IEC unit names. |